Project: Worked examples
Title: Thick-walled cylinder (Lamé)
Status: DRAFT · Author: TempestOS (2026-10-23) · Checker: — (—) · Issued: —

1. Scope and method

Calculates the radial, hoop and axial stresses at the bore and outer surface of a thick-walled cylinder under internal and external pressure. It also gives von Mises and Tresca stresses, the utilisation and a thin-wall estimate.

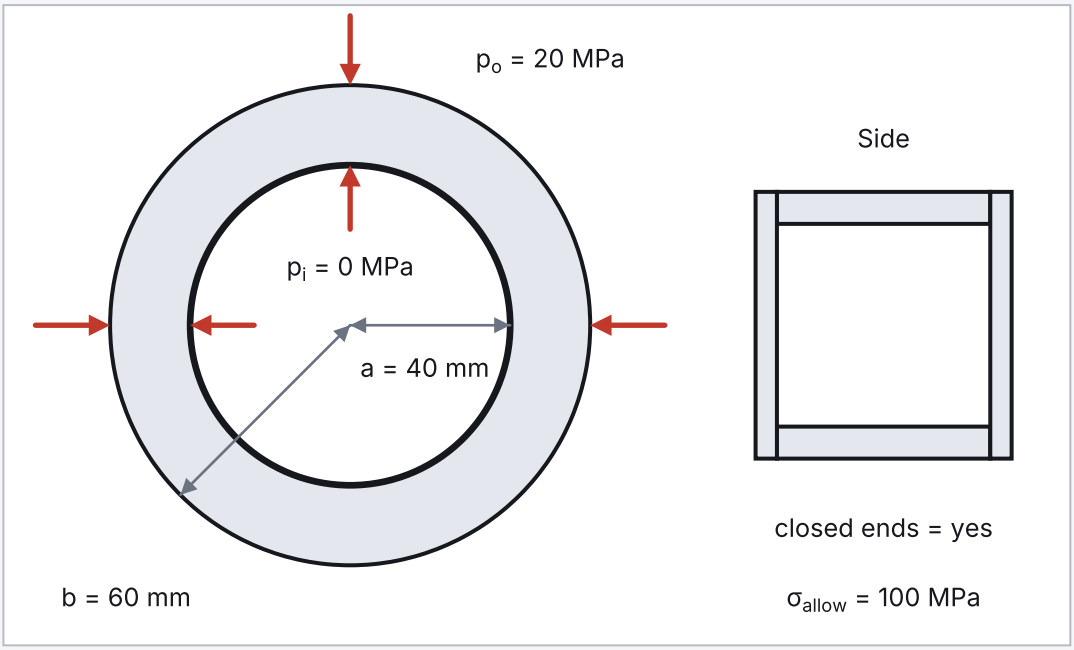


Figure 1 — A thick-walled cylinder in section, bore radius a, outer radius b, pressure inside and out; closed ends carry the pressures axially. Not to scale; the labels are the inputs as entered.

WHAT IT CALCULATES	RELATION
Lamé constants from the bore and outside pressures	$A = (a^2 p_i - b^2 p_o)/(b^2 - a^2)$; $B = a^2 b^2 (p_i - p_o)/(b^2 - a^2)$
Radial and hoop stress at radius r	$\sigma_r = A - B/r^2$; $\sigma_\theta = A + B/r^2$
Axial stress, closed or open ends	$\sigma_z = A$ (closed); $\sigma_z = 0$ (open)
Equivalent stresses from the three principal stresses	von Mises; Tresca = the largest principal difference

Basis: Thick-walled cylinders[1](Ch. 13, Table 13.5, 8th ed.); Stresses in pressurised cylinders[2](§3-14, 10th ed.).

2. Inputs

SYMBOL	DESCRIPTION	VALUE	UNIT	SOURCE	VALID RANGE
closed	Closed ends (axial stress carried by the wall)	yes		drawing: DRG-CYL-02 rev B	any
a	Inner radius	40	mm	drawing: DRG-CYL-02 rev B	0.1 … 5000 mm

SYMBOL	DESCRIPTION	VALUE	UNIT	SOURCE	VALID RANGE
b	Outer radius	60	mm	drawing: DRG-CYL-02 rev B	0.1 … 5000 mm
E	Young's modulus (the elastic collapse pressure under external pressure only)	207	GPa	assumption: Generic steel Young modulus for the elastic collapse pressure; take it from the material record of the tube.	1 … 500 GPa
v	Poisson's ratio (the elastic collapse pressure under external pressure only)	0.3		assumption: Generic steel Poisson ratio for the elastic collapse pressure; take it from the material record of the tube.	0.1 … 0.49
p_i	Internal pressure	0	MPa	assumption: Vented bore.	0 … 1000 MPa
p_o	External pressure	20	MPa	spec: SPEC-SUB-1 §2 external pressure	0 … 1000 MPa
σ_{allow}	Allowable stress	100	MPa	spec: SPEC-SUB-1 §4	1 … 3000 MPa

3. Assumptions

A1. The cylinder is long, elastic and isotropic, and its stresses vary with radius only. (Lamé solution, with end effects neglected.)
A2. Closed ends carry the pressure end load as a uniform axial stress. (σ_z = A.)
A3. E (Young's modulus (the elastic collapse pressure under external pressure only)) is taken as 207 GPa. (Generic steel Young modulus for the elastic collapse pressure; take it from the material record of the tube.)
A4. v (Poisson's ratio (the elastic collapse pressure under external pressure only)) is taken as 0.3. (Generic steel Poisson ratio for the elastic collapse pressure; take it from the material record of the tube.)
A5. p_i (Internal pressure) is taken as 0 MPa. (Vented bore.)

4. Working

[9] A = (a^2 * p_i - b^2 * p_o) / (b^2 - a^2) [1](Table 13.5)
= ((40 mm)^2 × 0 MPa − (60 mm)^2 × 20 MPa) / ((60 mm)^2 − (40 mm)^2)
= -36 MPa
Lamé constant A

[10] B = a^2 * b^2 * (p_i - p_o) / (b^2 - a^2) [1](Table 13.5)
= (40 mm)^2 × (60 mm)^2 × (0 MPa − 20 MPa) / ((60 mm)^2 − (40 mm)^2)
= -57600 MPa·mm²
Lamé constant B

[11] $\sigma_{r,a}$ = A - B / a^2
= (-36 MPa) − (-57600 MPa·mm²) / (40 mm)^2
= 0 MPa
Radial stress at the bore

[12] $\sigma_{t,a}$ = A + B / a^2
= -36 MPa + (-57600 MPa·mm²) / (40 mm)^2
= -72 MPa

Hoop stress at the bore

[13] $\sigma_{r,b}$ = A - B / b^2
= (-36 MPa) − (-57600 MPa·mm²) / (60 mm)^2
= -20 MPa

Radial stress at the outer surface

[14] $\sigma_{t,b}$ = A + B / b^2
= -36 MPa + (-57600 MPa·mm²) / (60 mm)^2
= -52 MPa

Hoop stress at the outer surface

[15] σ_z = A
= -36 MPa
= -36 MPa

Axial stress (closed ends)

[16] $\sigma_{vm,a}$ = sqrt(((($\sigma_{t,a}$ - $\sigma_{r,a}$)^2 + ($\sigma_{r,a}$ - σ_z)^2 + (σ_z - $\sigma_{t,a}$)^2) / 2)
= sqrt((((-72 MPa) − 0 MPa)^2 + (0 MPa − (-36 MPa))^2 + ((-36 MPa) − (-72 MPa))^2) / 2)
= 62.354 MPa

von Mises stress at the bore

[17] $\sigma_{vm,b}$ = sqrt(((($\sigma_{t,b}$ - $\sigma_{r,b}$)^2 + ($\sigma_{r,b}$ - σ_z)^2 + (σ_z - $\sigma_{t,b}$)^2) / 2)
= sqrt((((-52 MPa) − (-20 MPa))^2 + ((-20 MPa) − (-36 MPa))^2 + ((-36 MPa) − (-52 MPa))^2) / 2)
= 27.713 MPa

von Mises stress at the outer surface

[18] $\sigma_{tr,a}$ = max($\sigma_{t,a}$, $\sigma_{r,a}$, σ_z) - min($\sigma_{t,a}$, $\sigma_{r,a}$, σ_z)
= max(-72 MPa, 0 MPa, -36 MPa) − min(-72 MPa, 0 MPa, -36 MPa)
= 72 MPa

Tresca stress at the bore

[19] $\sigma_{tr,b}$ = max($\sigma_{t,b}$, $\sigma_{r,b}$, σ_z) - min($\sigma_{t,b}$, $\sigma_{r,b}$, σ_z)
= max(-52 MPa, -20 MPa, -36 MPa) − min(-52 MPa, -20 MPa, -36 MPa)
= 32 MPa

Tresca stress at the outer surface

[20] $\sigma_{vm,max}$ = max($\sigma_{vm,a}$, $\sigma_{vm,b}$)
= max(62.354 MPa, 27.713 MPa)
= 62.354 MPa

Maximum von Mises stress

[21] u = $\sigma_{vm,max}$ / σ_{allow}
= 62.354 MPa / 100 MPa
= 0.62354

Utilisation

[22] σ_{thin} = (p_i - p_o) * (a + b) / 2 / (b - a)
= (0 MPa − 20 MPa) × (40 mm + 60 mm) / 2 / (60 mm − 40 mm)
= -50 MPa

Thin-wall hoop estimate at the mean radius

[23] k_{wall} = b / a
= 60 mm / 40 mm
= 1.5

Wall ratio b/a

5. Results

SYMBOL	VALUE	DESCRIPTION
$\sigma_{t,a}$	-72 MPa	Hoop stress, bore
$\sigma_{vm,max}$	62.35 MPa	Maximum von Mises (governing)
$\sigma_{r,a}$	0 MPa	Radial stress, bore
$\sigma_{r,b}$	-20 MPa	Radial stress, outer
$\sigma_{t,b}$	-52 MPa	Hoop stress, outer
σ_z	-36 MPa	Axial stress
$\sigma_{vm,a}$	62.35 MPa	von Mises, bore
$\sigma_{vm,b}$	27.71 MPa	von Mises, outer
$\sigma_{tr,a}$	72 MPa	Tresca, bore
$\sigma_{tr,b}$	32 MPa	Tresca, outer
σ_{thin}	-50 MPa	Thin-wall hoop estimate
k_{wall}	1.5	Wall ratio
u	0.62	Utilisation

6. Checks (governing first)

RESULT	CHECK	DEMAND	CAPACITY	UTILISATION	REF
PASS	von Mises against allowable	62.35 MPa	100 MPa	0.62	

7. Applicability

None.

8. Verification

Template mech.thick-cylinder@2.2.1 Content hash e204fc736739a5e7 Engine calc-engine/1.0.0
Known-answer cases passed 3/3 Test run 2026-10-23 12:00 UTC TempestOS 0.26.0

9. References

1. Young WC, Budynas RG, Sadegh AM. Roark's Formulas for Stress and Strain. 8th ed. McGraw-Hill; 2012.
2. Budynas RG, Nisbett JK. Shigley's Mechanical Engineering Design. 11th ed. McGraw-Hill Education; 2020.